Signals and Systems (Tutorial)

Topic - System Properties **Week-21-25th August, 2023**

Q1. Compute and plot y[n] = x[n] * h[n], where

$$x(n) = 1\,, \quad 3 \le n \le 8$$

$$0, \quad \text{Otherwise}$$

$$h(n) = 1\,, \quad 4 \le n \le 15$$

$$0, \quad \text{Otherwise}$$

Q2. Determine and sketch the convolution of the following two signals:

$$x(t) = \begin{cases} t+1\,, & 0 \le t \le 1 \\ 2-t\,, & 1 < t \le 2 \\ 0\,, & \text{Otherwise} \end{cases}$$

$$h(t) = \delta(t+2) + 2\delta(t+1).$$

Q3. A linear system S has the relationship

$y[n] = \sum_{k=-\infty}^{+\infty} x[k]g[n-2k]$ **, between its input x[n] and output y[n].**

Where g[n] = u[n]- u[n-4].

$(a).\ Determine\ y[n]\ when\ x[n] = \delta[n-1].$

$(b).\ Determine\ y[n]\ when\ x[n] = \delta[n-2].$

$(c).\ Is\ S\ an\ LTI\ ?\ .$

$(d).\ Determine\ y[n]\ when\ x[n] = u[n].$

Q4. Consider a discrete time system S1 with impulse response

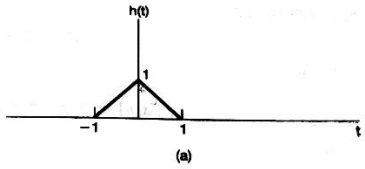
$$h(n) = \left(\frac{1}{5}\right)^n u[n].$$

$(a)\ Find\ the\ integer\ A\ such\ that\ h[n] - Ah[n-1] = \delta[n].$

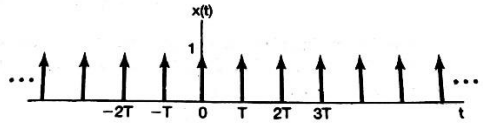
$(b)\ Using\ the\ result\ from\ part\ (a), determine\ the\ impulse\ response$

$g[n]\ of\ an\ LTI\ system\ S2\ , which\ is\ the\ inverese\ system\ of\ S1.$

Q5. Let h(t) be the triangular pulse and x(t) be the impulse train shown in the figure (a) and (b).



(a)



(b)

$$\textit{that is } x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT),$$

$$\textit{Determine and sketch } y(t) = x(t) * h(t) \textit{for thr followingh values of } T:$$

$$(a) T = 4 \;\; (b) T = 2 \;\; (c) T = \frac{3}{2} \quad (d)\; T = 1$$

Q6. Following are the impulse responses of Discrete-time LTI systems. Determine wether each system is causal and/or stable. Justify your answers.

$$(a)\; h(n) = \left(\frac{1}{5}\right)^n u[n] \;\; , \;\; \textbf{(b)}\; h(n) = (0.8)^n u[n+2]$$

$$(c) \left(-\frac{1}{2}\right)^n u[n] + (1.01)^n u[n-1] \,, \;\; (d) h[n] = n\left(\frac{1}{3}\right)^n u[n-1]$$

Q7. Following are the impulse responses of Continuous-time LTI systems. Determine wether each system is causal and/or stable. Justify your answers.

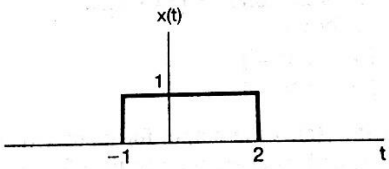
$$(a)\; h(t) = e^{-4t} u(t-2)\,, \qquad (b)\; h(t) = e^{-6t} u(3-t), \qquad (c) h(t) = e^{2t} u(-1-t)$$

Q8. (a)Consider an LTI system with the input and output related through the equation

$$y(t) = \int_{-\infty}^{t} e^{-(t-\tau)} x(\tau - 2) d\tau$$

What is the impulse response h(t) for this System?

(b) Compute the response of the above system if the input x(t) is as shown in the figure.



Q.9 $Suppose\ the\ signal\ x(t) = u(t+0.5) - u(t-0.5) and\ the\ signal$

$$h(t) = e^{j\omega_o t}$$

$$(a) Determine\ a\ value\ of\ \omega_o\ which\ ensures\ that\ y(0) = 0,$$

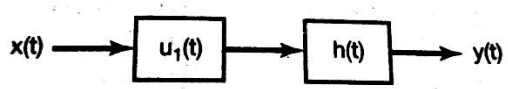
$$where\ y(t) = x(t) * h(t)$$

$$(b) is\ your\ answer\ to\ the\ previous\ part\ unique\ ?$$

Q10. (a) Show that if the response of an LTI system to x(t) is the output y(t), then the response of the system to $x'(t) = \frac{dx(t)}{dt}$ **is** $y'(t)$.

Do this problem in three different ways:

- **(i)** **Directly from the property of linearity and time invariance and the fact that** $x'(t) = \lim_{h\to 0} \frac{x(t)-x(t-h)}{h}$
- **(ii)** **By differentiating the convolution integral.**
- **(iii)** **By examining the system given in figure below**



(b) Demonstrate the validity of the following relationships:

(i) $y'(t) = x(t) * h'(t)$

(ii) $y(t) = \left(\int_{-\infty}^{t} x(\tau)d\tau\right) * h'(t) = \int_{-\infty}^{t} [x'(\tau) * h(\tau)]d\tau = x'(t) * \left(\int_{-\infty}^{t} h(\tau)d\tau\right)$

(c) An LTI system has the response y(t)=sin($\omega_0 t$) to input x(t)= $e^{-5t}u(t)$**. Use the result of part (a) to aid in determining the impulse response of this system.**

(d) Let s(t) be the unit step response of a continuous time LTI system. Use part (b) to deduce that the response y(t) to the input x(t) is

$$y(t) = \left(\int_{-\infty}^{\infty} x'(\tau) * s(t-\tau)d\tau\right)\ show\ also\ that\ x(t) = \int_{-\infty}^{\infty} x'(\tau)u(t-\tau)d\tau$$

(e) Use the equation $y(t) = \left(\int_{-\infty}^{\infty} x'(\tau) * s(t-\tau)d\tau\right)$ **to determine the response of an LTI system with step response** $s(t) = (e^{-3t} - 2e^{-2t} + 1)u(t)$ **to the input**

x(t)= $e^{t}u(t)$.

Q11. Determine wether each of the following statements concerning LTI systems is true or false. Justify your answers.

- **(a) If h(t) is the impulse response of an LTI system and h(t) is periodic and nonzero, the system is unstable.**

(b) **The inverse of a causal LTI system is always causal.**

(c) **) If $h[n] \leq K$ for each n, where K is a given number, then the LTI system with h[n] as its impulse response is stable.**

(d) **If a discrete-time LTI system has an impulse response h[n] of finite duration, the system is stable.**